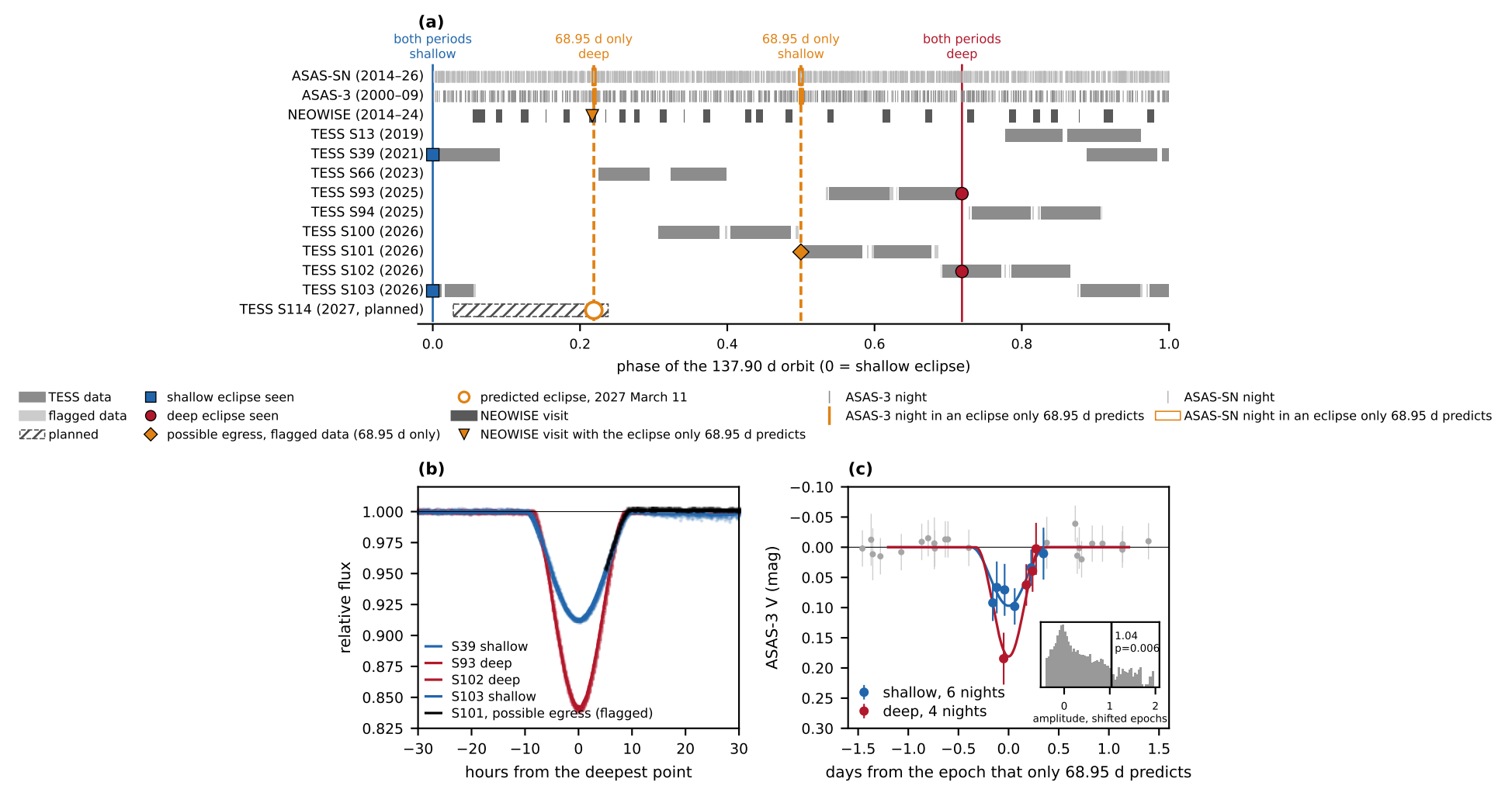

(a)
both periods shallow
68.95 d only deep
68.95 d only shallow
both periods deep
ASAS-SN (2014–26)
ASAS-3 (2000–09)
NEOWISE (2014–24)
TESS S13 (2019)
TESS S39 (2021)
TESS S66 (2023)
TESS S93 (2025)
TESS S94 (2025)
TESS S100 (2026)
TESS S101 (2026)
TESS S102 (2026)
TESS S103 (2026)
TESS S114 (2027, planned)
0.0
0.2
0.4
0.6
0.8
1.0
phase of the 137.90 d orbit (0 = shallow eclipse)
TESS data
flagged data
planned
shallow eclipse seen
deep eclipse seen
possible egress, flagged data (68.95 d only)
predicted eclipse, 2027 March 11
NEOWISE visit
NEOWISE visit with the eclipse only 68.95 d predicts
ASAS-3 night
ASAS-3 night in an eclipse only 68.95 d predicts
ASAS-SN night
ASAS-SN night in an eclipse only 68.95 d predicts
(b)
relative flux
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.825
S39 shallow
S93 deep
S102 deep
S103 shallow
S101, possible egress (flagged)
−30
−20
−10
0
10
20
30
hours from the deepest point
(c)
ASAS-3 V (mag)
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
0.25
0.30
shallow, 6 nights
deep, 4 nights
1.04
p=0.006
0
1
2
amplitude, shifted epochs
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
days from the epoch that only 68.95 d predicts